

The Activation Function and the Update Rule

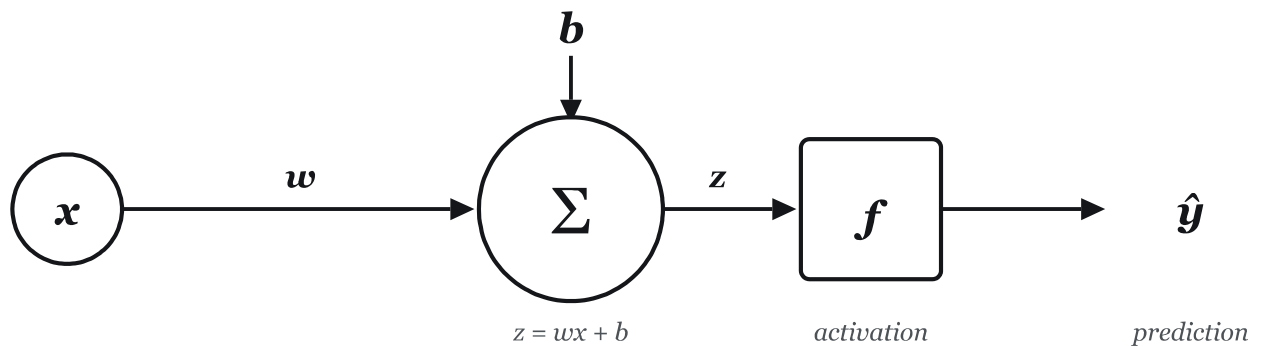
A single perceptron. All work by hand.

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PART 1 WHAT AN ACTIVATION FUNCTION DOES

1.1 The perceptron

FIGURE 1 – A SINGLE PERCEPTRON



The perceptron does two things in order. First arithmetic, producing z . Then the activation function f , producing the prediction.

$$z = wx + b \qquad \hat{y} = f(z)$$

The first part is **arithmetic**. The second part, f , is the **activation function**.

1.2 Why f is needed

Example. A perceptron decides whether a student passes. The input x is the marks obtained. Take $w = 1$ and $b = -40$.

Student	x	$z = wx + b$
A	65	$z = (1)(65) + (-40) = 25$
B	32	$z = (1)(32) + (-40) = -8$

The question asked was *pass or fail*. The required answer is 1 or 0. The arithmetic produced 25 and -8.

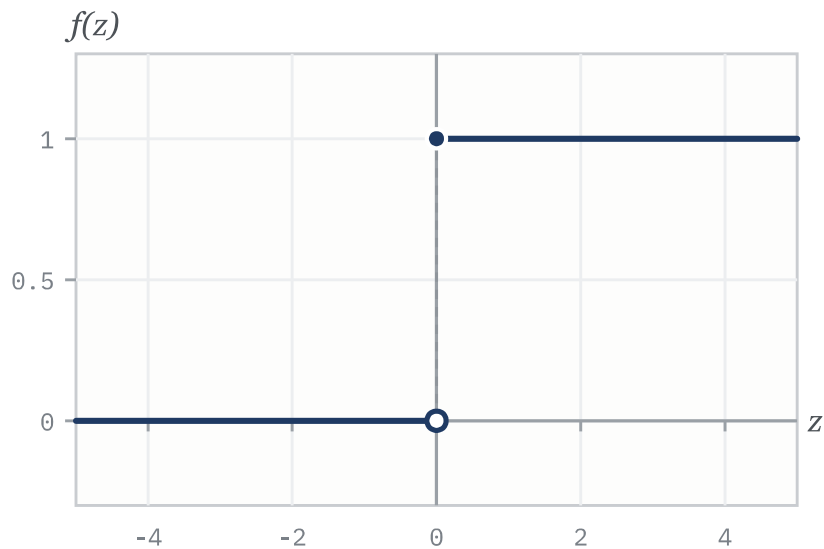
25 is not an answer to the question.

Something must convert z into the answer. That is f .

1.3 The threshold (step) function

$$f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

FIGURE 2 – THE THRESHOLD FUNCTION



The open circle at $(0, 0)$ is excluded; the filled dot at $(0, 1)$ is included. The function jumps at $z = 0$.

Apply it to the two students:

Student	z	$\hat{y} = f(z)$	Decision
A	25	$f(25) = 1$	pass
B	-8	$f(-8) = 0$	fail

1.4 The role of f

For a single perceptron, the activation function f : (i) takes z , which may be any real number, (ii) returns an output in a fixed range, and (iii) so lets the perceptron give an **answer** rather than a measurement.

2.1 Definitions

$$z = wx + b \quad \hat{y} = f(z) \quad L = \frac{1}{2}(\hat{y} - y)^2$$

y is the target, L is the loss, and α is the learning rate.

2.2 The rule

A large L means a bad prediction. We want L to be **smaller**, so we must change w in whichever direction reduces L . The gradient $\partial L / \partial w$ points in the direction in which L **increases**, so we move against it. This gives the **gradient descent update rule**:

$$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial L}{\partial w}$$

2.3 Find $\partial L / \partial w$

w affects z ; z affects $\hat{y}$; $\hat{y}$ affects L . Apply the chain rule.

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w}$$

Complete each factor.

Since $L = \frac{1}{2}(\hat{y} - y)^2$ and y is a constant

$$\frac{\partial L}{\partial \hat{y}} = \underline{\hspace{2cm}}$$

Since $\hat{y} = f(z)$

$$\frac{\partial \hat{y}}{\partial z} = \underline{\hspace{2cm}}$$

Since $z = wx + b$, and x and b are constants

$$\frac{\partial z}{\partial w} = \underline{\hspace{2cm}}$$

Multiply the three factors

$$\frac{\partial L}{\partial w} = \underline{\hspace{2cm}}$$

2.4 Substitute into the rule

$$w_{\text{new}} = \underline{\hspace{2cm}}$$

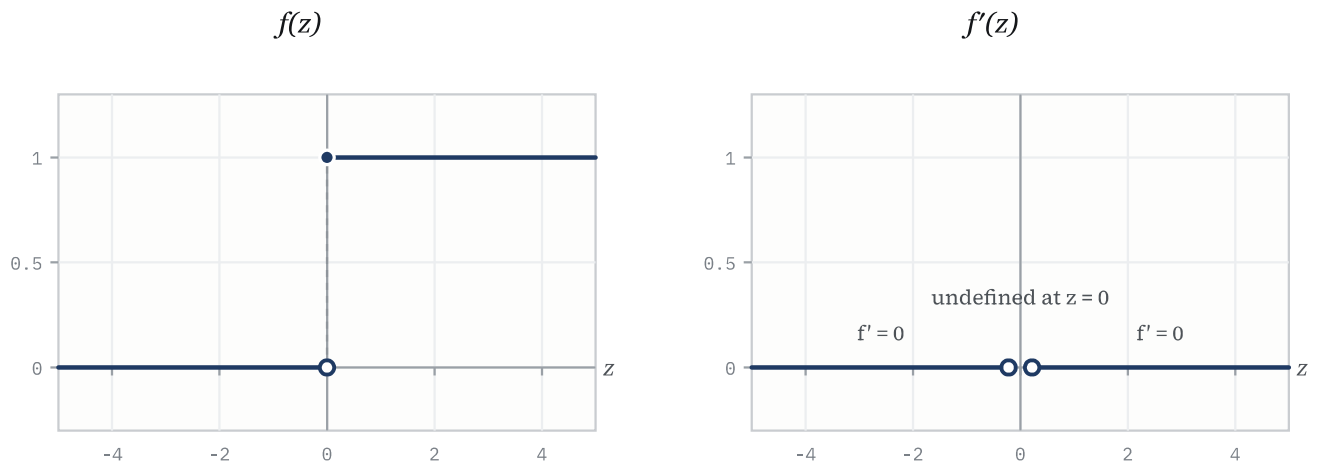
Check your answer against the box below. Use this form for every numerical from here on.

$$w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

Three factors control the size of the step: the **error** $(\hat{y} - y)$, the **input** x , and the **derivative of the activation function** $f'(z)$.

3.1 Its derivative

FIGURE 3 – THE THRESHOLD FUNCTION AND ITS DERIVATIVE



Both plots use the same vertical scale. The derivative is flat on zero across the whole width. At $z = 0$ the function jumps, so the derivative does not exist there.

For $z > 0$, $f(z) = 1$, a constant

$$f'(z) = \underline{\hspace{2cm}}$$

For $z < 0$, $f(z) = 0$, a constant

$$f'(z) = \underline{\hspace{2cm}}$$

At $z = 0$ the function jumps, so $f'(0)$ is **undefined**. Therefore:

$$\therefore f'(z) = 0 \quad \text{for all } z \neq 0$$

3.2 Numerical

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 2 \quad w = 1 \quad b = 0 \quad y = 0 \quad \alpha = 0.5$

Weighted sum

$$z = wx + b = (1)(2) + 0 = 2$$

Activation

$$\hat{y} = f(2) = 1 \quad (\text{since } 2 \geq 0)$$

Error

$$\hat{y} - y = 1 - 0 = 1$$

Derivative

$$f'(2) = 0$$

Gradient

$$\frac{\partial L}{\partial w} = (\hat{y} - y) f'(z) x = (1)(0)(2) = 0$$

Update

$$w_{\text{new}} = 1 - (0.5)(0) = 1$$

3.3 Conclusion

The error is 1, the largest it can be. The weight did not change.

The update is zero because

Increasing α would _____, because

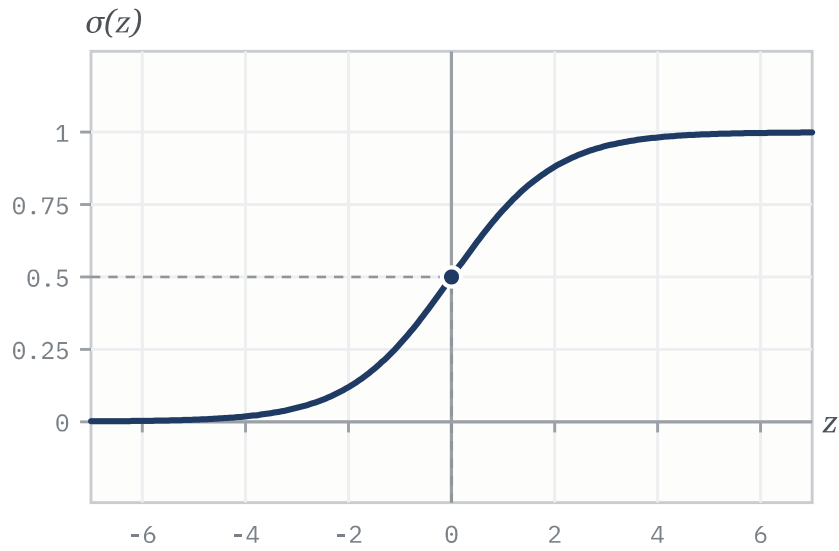
A usable activation function must have a non-zero derivative.

PART 4 THE SIGMOID FUNCTION

4.1 Definition

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

FIGURE 4 – THE SIGMOID FUNCTION



Compare with Figure 2. The sigmoid is **smooth**, so it has a derivative everywhere. It passes through $(0, 0.5)$, and $0 < \sigma(z) < 1$ for every z .

4.2 Derive $\sigma'(z)$

Write $\sigma(z) = (1 + e^{-z})^{-1}$.

Step 1 (given). By the chain rule

$$\sigma'(z) = -(1 + e^{-z})^{-2} \cdot \frac{d}{dz}(1 + e^{-z})$$

Step 2. Evaluate the inner derivative

$$\frac{d}{dz}(1 + e^{-z}) = \underline{\hspace{2cm}}$$

Step 3. Substitute into Step 1 and simplify

$$\sigma'(z) = \underline{\hspace{2cm}}$$

Step 4 (given). Rewrite the numerator using $e^{-z} = (1 + e^{-z}) - 1$

$$\sigma'(z) = \frac{(1 + e^{-z}) - 1}{(1 + e^{-z})^2}$$

Step 5. Split into two fractions

$$\sigma'(z) = \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

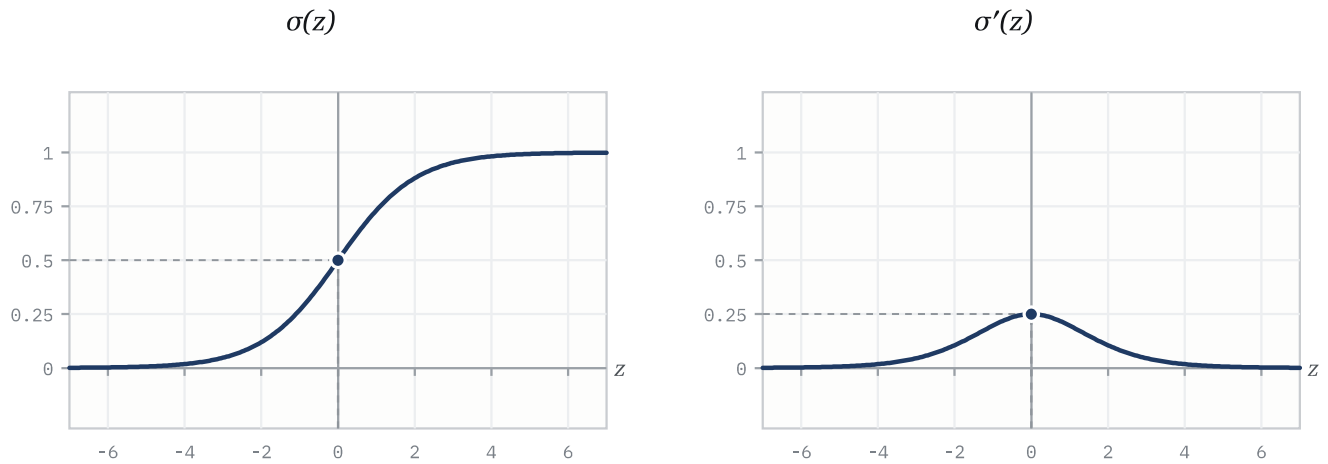
Step 6. Each term is a power of $\sigma(z)$

$$\sigma'(z) = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\sigma'(z) = \sigma(z) (1 - \sigma(z))$$

4.3 The two graphs

FIGURE 5 – THE SIGMOID AND ITS DERIVATIVE



Both plots use the same vertical scale, 0 to 1. Note where σ' is large and where it is small. Unlike the threshold function, $\sigma'(z)$ is never zero.

PART 5 WORKED EXAMPLE: THE WEIGHT MOVES THE CORRECT WAY

The same numbers as Part 3. **Only f has changed.**

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 2$ $w = 1$ $b = 0$ $y = 0$ $\alpha = 0.5$

Step 1. Weighted sum

$$z = wx + b = (1)(2) + 0 = 2$$

Step 2. Activation

$$\hat{y} = \sigma(2) = \frac{1}{1 + e^{-2}} = \frac{1}{1 + 0.135335} = \frac{1}{1.135335} = 0.880797$$

Step 3. Error

$$\hat{y} - y = 0.880797 - 0 = 0.880797$$

Step 4. Derivative

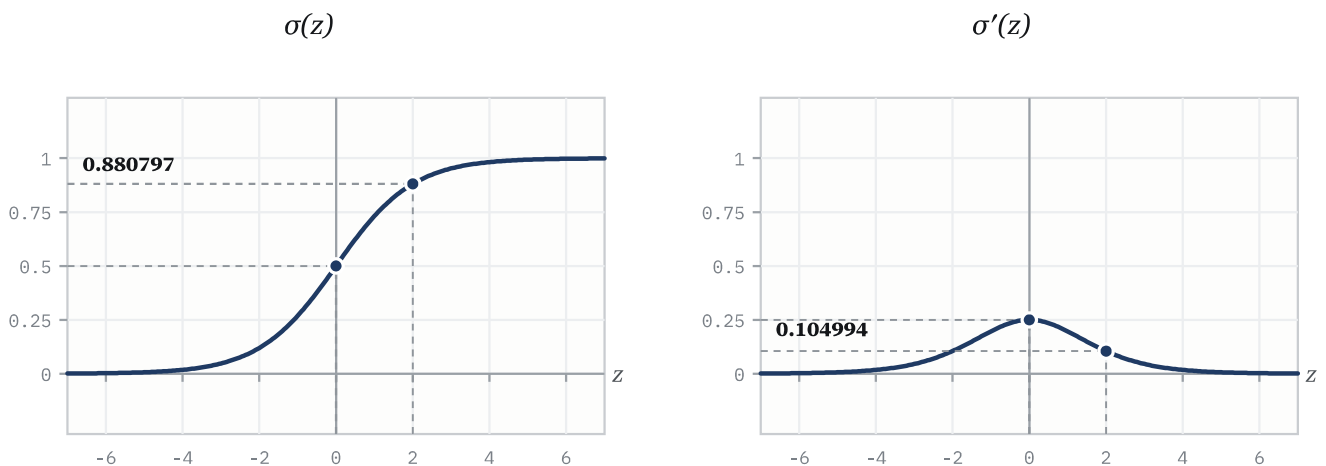
$$\sigma'(2) = \sigma(2)(1 - \sigma(2)) = (0.880797)(0.119203) = 0.104994$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = (0.880797)(0.104994)(2) = 0.184956$$

Step 6. Update

$$w_{\text{new}} = 1 - (0.5)(0.184956) = 1 - 0.092478 = 0.907522$$

FIGURE 6 — THE SAME TWO GRAPHS, WITH $z = 2$ MARKED

At $z = 2$ the sigmoid is already flattening, so $\sigma'(2) = 0.104994$ has fallen well below its peak. Read both values off the dashed guides.

Direction check

Requirement	Consequence
Target is $y = 0$	$\hat{y}$ must decrease
$\hat{y} = \sigma(z)$ increases with z	z must decrease
$z = wx + b$ with $x = 2 > 0$	w must decrease
Result	$w : 1 \rightarrow 0.907522 \quad \checkmark$

Comparison with Part 3

IDENTICAL x, w, b, y AND a . ONLY $f'(z)$ DIFFERS.

Activation	$f'(2)$	$\partial L / \partial w$	w_{new}
Threshold	0	0	1 (no change)
Sigmoid	0.104994	0.184956	0.907522

PART 6 PRACTICE

Show every step. Give all values to six decimal places.

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 3 \quad w = 0.5 \quad b = -1 \quad y = 1 \quad \alpha = 2$

USEFUL $e^{-0.5} = 0.606531$

Step 1. Weighted sum

$$z = wx + b = \underline{\hspace{4cm}}$$

Step 2. Activation

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} = \underline{\hspace{4cm}}$$

Step 3. Error

$$\hat{y} - y = \underline{\hspace{4cm}}$$

Step 4. Derivative

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) = \underline{\hspace{4cm}}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \sigma'(z) x = \underline{\hspace{4cm}}$$

Step 6. Update

$$w_{\text{new}} = \underline{\hspace{4cm}}$$

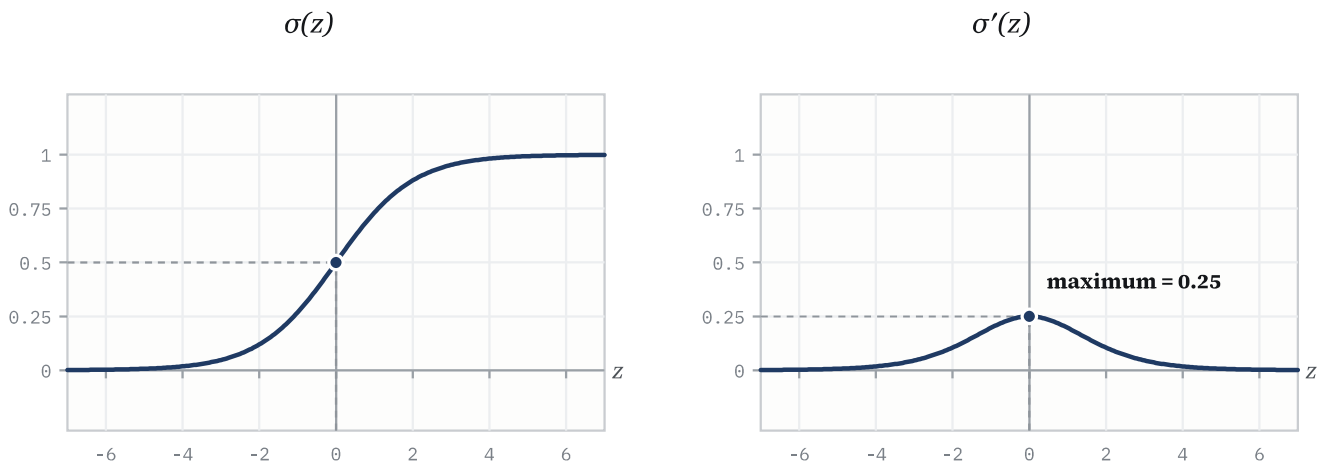
Step 7 — state the direction of the change and justify it in one line.

ANSWERS, FOR CHECKING ONLY

$z = 0.5$ | $\hat{y} = 0.622459$ | $\hat{y} - y = -0.377541$ | $\sigma'(z) = 0.235004$ | $\partial L / \partial w = -0.266170$ | w_{new}
 $= 1.032341$

7.1 Read the maximum off the curve

FIGURE 7 — KEY FIGURE. READ THE PEAK HEIGHT FROM THE RIGHT-HAND PLOT



Both vertical axes run from 0 to 1 with identical marks, so the two curves can be compared directly. The dashed guides on the right-hand plot locate the peak.

Fill in by reading Figure 7.

The peak of $\sigma'(z)$ occurs at

$$z = \underline{\hspace{2cm}}$$

The height of that peak is

$$\sigma'_{\max} = \underline{\hspace{2cm}}$$

On the left-hand plot, the value of $\sigma(z)$ at that same z is

$$\sigma(z) = \underline{\hspace{2cm}}$$

$$\sigma'(z) \leq 0.25 \quad \text{for every } z$$

Confirm with the formula: at $z = 0$ we have $\sigma(0) = 0.5$, so $\sigma'(0) = (0.5)(1 - 0.5) = 0.25$. ✓

7.2 The problem

At **every** value of z , the sigmoid multiplies the gradient by at most 0.25. The numerical below shows what that costs even in the best possible case, where $z = 0$ and the derivative takes its largest value.

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 1$ $w = 0$ $b = 0$ $y = 1$ $\alpha = 1$

Weighted sum

$$z = wx + b = (0)(1) + 0 = 0$$

Activation

$$\hat{y} = \sigma(0) = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5$$

Error

$$\hat{y} - y = 0.5 - 1 = -0.5$$

Derivative

$$\sigma'(0) = (0.5)(1 - 0.5) = 0.25 \quad \leftarrow \text{the largest value it can ever take}$$

Gradient

$$\frac{\partial L}{\partial w} = (-0.5)(0.25)(1) = -0.125$$

Update

$$w_{\text{new}} = 0 - (1)(-0.125) = 0.125$$

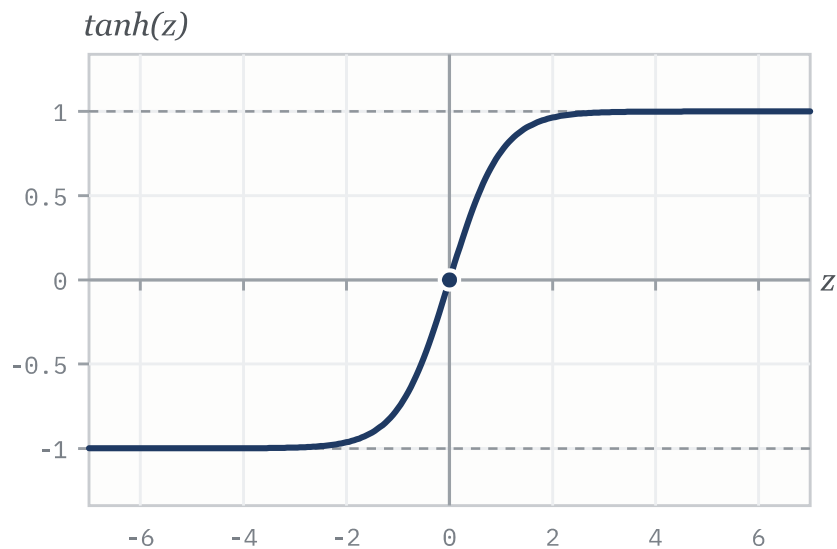
This is the sigmoid at its **best possible** point, and it still shrinks the gradient to one quarter. The step taken is only 0.125.

PART 8 THE TANH FUNCTION

8.1 Definition

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad -1 < \tanh(z) < 1$$

FIGURE 8 – THE TANH FUNCTION



Compare with Figure 4. The same S-shape, but the output range is -1 to 1 , and the curve passes through the origin.

8.2 Its derivative

Let $u = e^z - e^{-z}$ and $v = e^z + e^{-z}$, so $\tanh = u/v$.

Step 1 (given). Differentiate each part

$$u' = e^z + e^{-z} = v \quad v' = e^z - e^{-z} = u$$

Step 2. Quotient rule, then substitute $u' = v$ and $v' = u$

$$\tanh'(z) = \frac{u'v - uv'}{v^2} = \underline{\hspace{2cm}}$$

Step 3. Split the fraction and simplify

$$\tanh'(z) = \underline{\hspace{2cm}}$$

The two parts exchange with each other. That is what makes this derivation short.

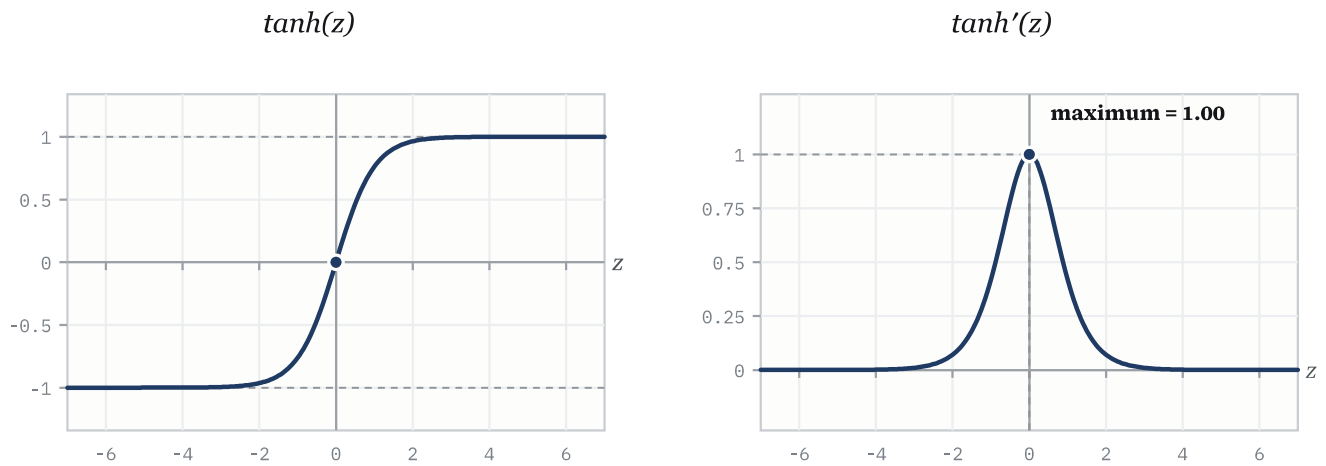
$$\tanh'(z) = 1 - \tanh^2(z)$$

At $z = 0$, since $\tanh(0) = 0$

$$\tanh'(0) = 1 - (0)^2 = \underline{\hspace{2cm}}$$

8.3 The two graphs

FIGURE 9 – TANH AND ITS DERIVATIVE



The derivative reaches a peak of 1.00 at $z=0$, compared with 0.25 for the sigmoid in Figure 7. Its vertical axis runs 0 to 1, the same as Figure 7, so the two peaks are directly comparable.

8.4 The same numerical, with tanh

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN, UNCHANGED FROM 7.2 $x = 1$ $w = 0$ $b = 0$ $y = 1$ $\alpha = 1$

Weighted sum

$$z = wx + b = (0)(1) + 0 = 0$$

Activation

$$\hat{y} = \tanh(0) = 0$$

Error

$$\hat{y} - y = 0 - 1 = -1$$

Derivative

$$\tanh'(0) = 1 - (0)^2 = 1$$

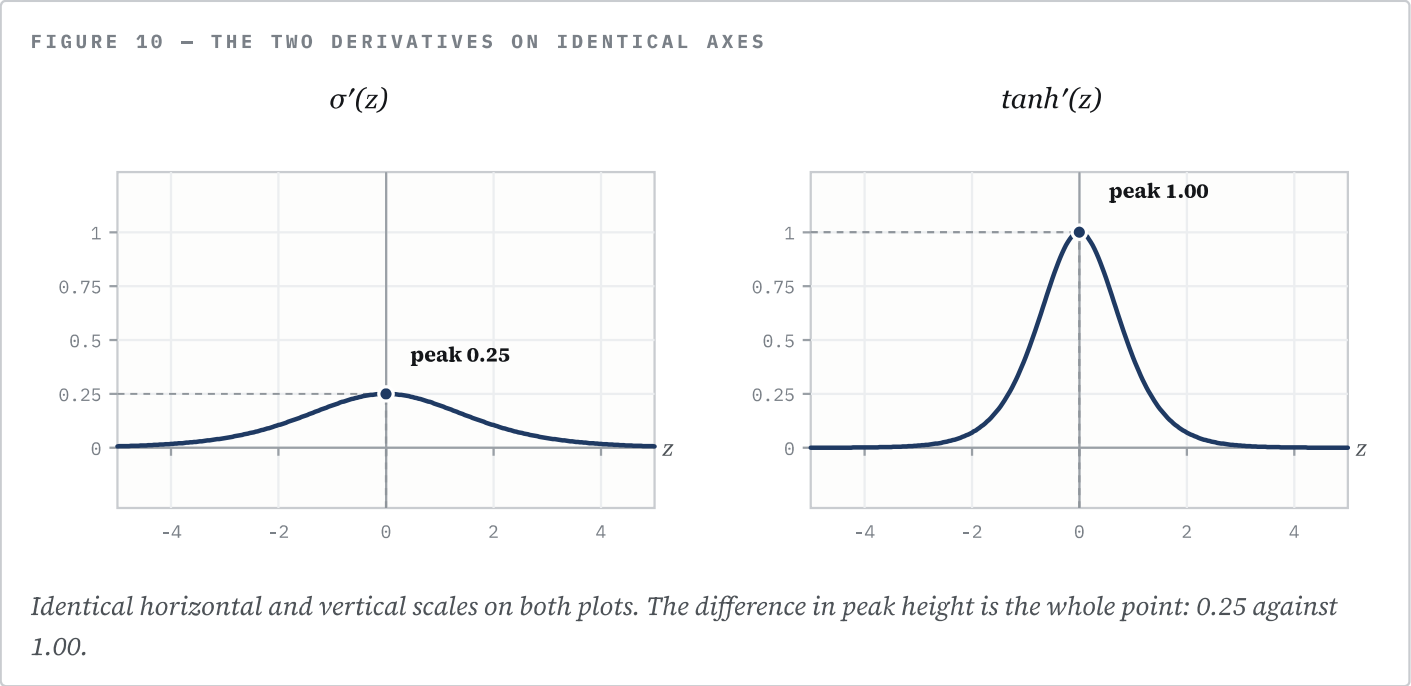
Gradient

$$\frac{\partial L}{\partial w} = (-1)(1)(1) = -1$$

Update

$$w_{\text{new}} = 0 - (1)(-1) = 1$$

8.5 Comparison



	$\hat{y}$	$\hat{y} - y$	$f'(0)$	$\partial L / \partial w$	$w : 0 \rightarrow$	Step size
Sigmoid	0.5	-0.5	0.25	-0.125	0.125	0.125
Tanh	0	-1	1	-1	1	1

$$\frac{\text{tanh step}}{\text{sigmoid step}} = \frac{1}{0.125} = 8$$

Account for the factor of 8. Two quantities changed.

$$\begin{aligned} \text{error ratio} &= \frac{1}{0.5} = \underline{\hspace{2cm}} & \text{derivative ratio} &= \frac{1}{0.25} = \underline{\hspace{2cm}} \\ & & \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} &= 8 \end{aligned}$$

SUMMARY THE THREE ACTIVATION FUNCTIONS

Activation	$f(z)$	$f'(z)$	$\max f'$	Learns?
Threshold	1 if $z \geq 0$, else 0	0	0	No
Sigmoid	$1 / (1 + e^{-z})$	$\sigma(z)(1 - \sigma(z))$	0.25	Yes, slowly
Tanh	$(e^z - e^{-z}) / (e^z + e^{-z})$	$1 - \tanh^2(z)$	1	Yes, faster

In the update rule

$$w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

the factor $f'(z)$ decides **whether** learning happens at all, and **how fast**.

In one line: why can a perceptron using the threshold function never be trained by this rule?

All numerical values in this worksheet are exact to six decimal places. Every plot is drawn to scale, and all derivative plots share a vertical axis running from 0 to 1 so that peak heights may be compared directly.